Midterm Exam

- Your exam time
  - See email
- Your room & seat assignments, exam instructions
  - TBA

- We will open the computer rooms 6-9:30pm next Mon/Tue, so you can familiarize yourself with the exam computer system

Course schedule

- Three parts:
  - Python Programming & AI (9 weeks)
    - Basic principles of computer programming
    - Introduction of AI
  - Electronic Engineering (3 weeks) 教学中心103
    - Fundamental semiconductor components of the information age
    - How electronic technology achieves computing and communication with the physical world;
    - Typical systems based on electronic technology.
  - Course Projects (4 weeks) TBA. 1-2 lectures and then no lecture.
    - 4-5 themes focusing on PP, AI, or EE aspects.
    - Choose one of the topics in groups and conduct a 4-week course project

Language Models

Outline

- Large language models (LLM)
  - Since 2022

3

LLM is a special case of PLM

- Pretrained language models (PLM)
  - Since 2018

2

PLM is a special case of LM

- Language models (LM)
  - Since 1950?

1

Language Modeling

- **Goal:** compute the probability of a sentence (sequence of words)

$$P(w_1, w_2, w_3, \cdots, w_n)$$

- The Chain Rule applied to compute joint probability of words in sentence

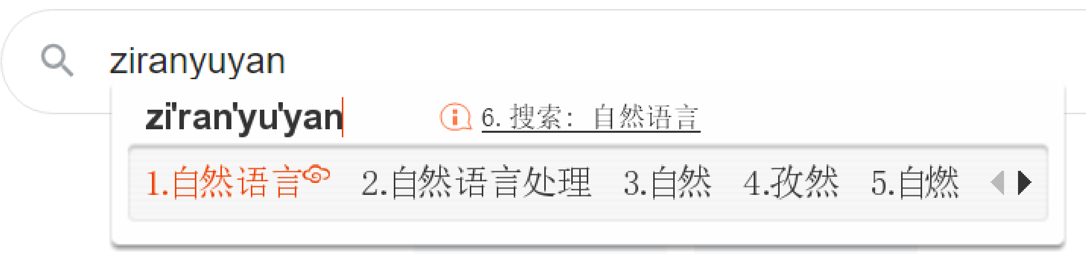
$$P(w_1, w_2, \dots, w_n) = \prod_i P(w_i | w_1, w_2, \dots, w_{i-1})$$

- Ex.

$$P(\text{its water is so transparent}) =$$
$$P(\text{its}) \times P(\text{water}|\text{its}) \times P(\text{is}|\text{its water}) \times$$
$$P(\text{so}|\text{its water is}) \times P(\text{transparent}|\text{its water is so})$$

Applications

- Original applications
  - Speech Recognition
    - $P(\text{I saw a van}) >> P(\text{eyes awe of an})$
  - Chinese IME
    - $P(\text{自然语言}) >> P(\text{孜然鱼雁})$
  - …



Text generation with LM

today the __________

condition on this

get probability distribution

company	0.153
bank	0.153
price	0.077
italian	0.039
emirate	0.039
…	

sample

Text generation with LM

$\underbrace{\textit{today the price}}_{\text{condition on this}}$ __________

get probability distribution

of	0.308
for	0.050
it	0.046
to	0.046
is	0.031
…	

sample

Text generation with LM

today the price of ________

condition on this

get probability distribution

the	0.072
18	0.043
oil	0.043
its	0.036
gold	0.018
…	

sample

Text generation with LM

today the price of gold _________

Text generation with LM

today the price of gold per ton , while production of shoe lasts and shoe industry , the bank intervened just after it considered and rejected an imf demand to rebuild depleted european stocks , sept 30 end primary 76 cts a share .

Main Challenge

- How to estimate these conditional probabilities?
- Could we rely on statistics from a corpus?

$$P(\text{the}|\text{its water is so transparent that}) = \frac{Count(\text{its water is so transparent that the})}{Count(\text{its water is so transparent that})}$$

- Will this work?
  - No! Impossible for a corpus to cover all possible texts

N-gram model (1950s, 1990s)

- We approximate each component in the product

$$P(w_i|w_1, w_2, \dots, w_{i-1}) \approx P(w_i|w_{i-k}, \dots, w_{i-1})$$

  - This is called Markov Assumption

- Ex:
  - Bigram model: condition on the previous word:

$$P(w_i|w_1, w_2, \dots, w_{i-1}) \approx P(w_i|w_{i-1})$$

  - Trigram model: condition on the previous two word:

$$P(w_i|w_1, w_2, \dots, w_{i-1}) \approx P(w_i|w_{i-2}, w_{i-1})$$

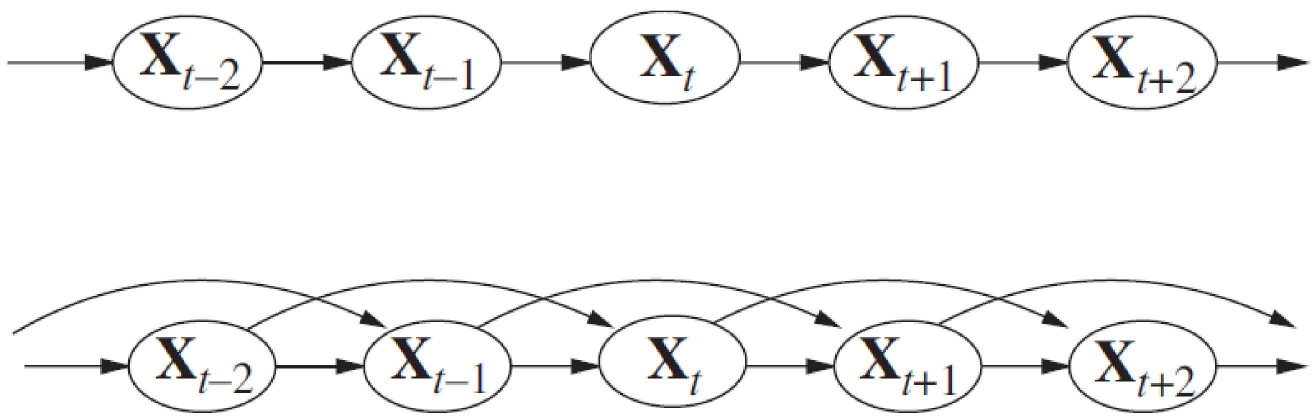
Text generation with a trigram model

today the price of gold per ton , while production of shoe lasts and shoe industry , the bank intervened just after it considered and rejected an imf demand to rebuild depleted european stocks , sept 30 end primary 76 cts a share .

- Surprisingly grammatical!
- …but **incoherent.** We need to consider more than three words at a time if we want to model language well.
- But increasing n worsens the sparsity problem, and increases model size…

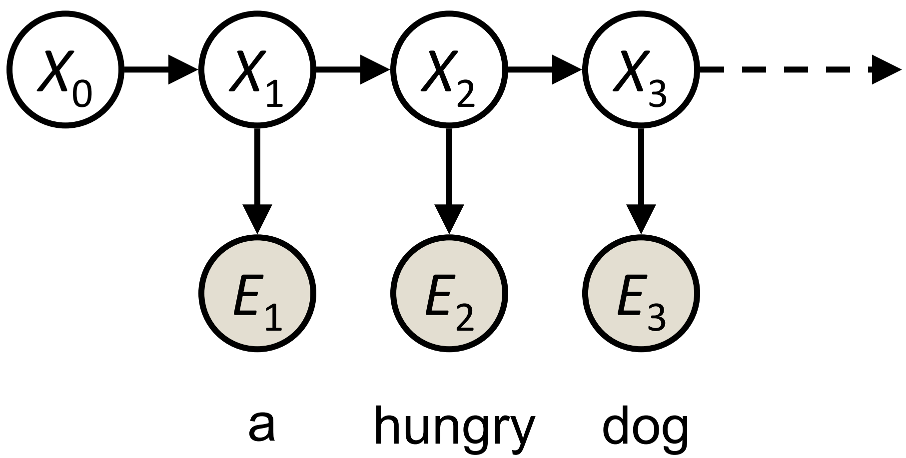
Markov Models (1910s)

- N-gram model = (N-1)-th order Markov model



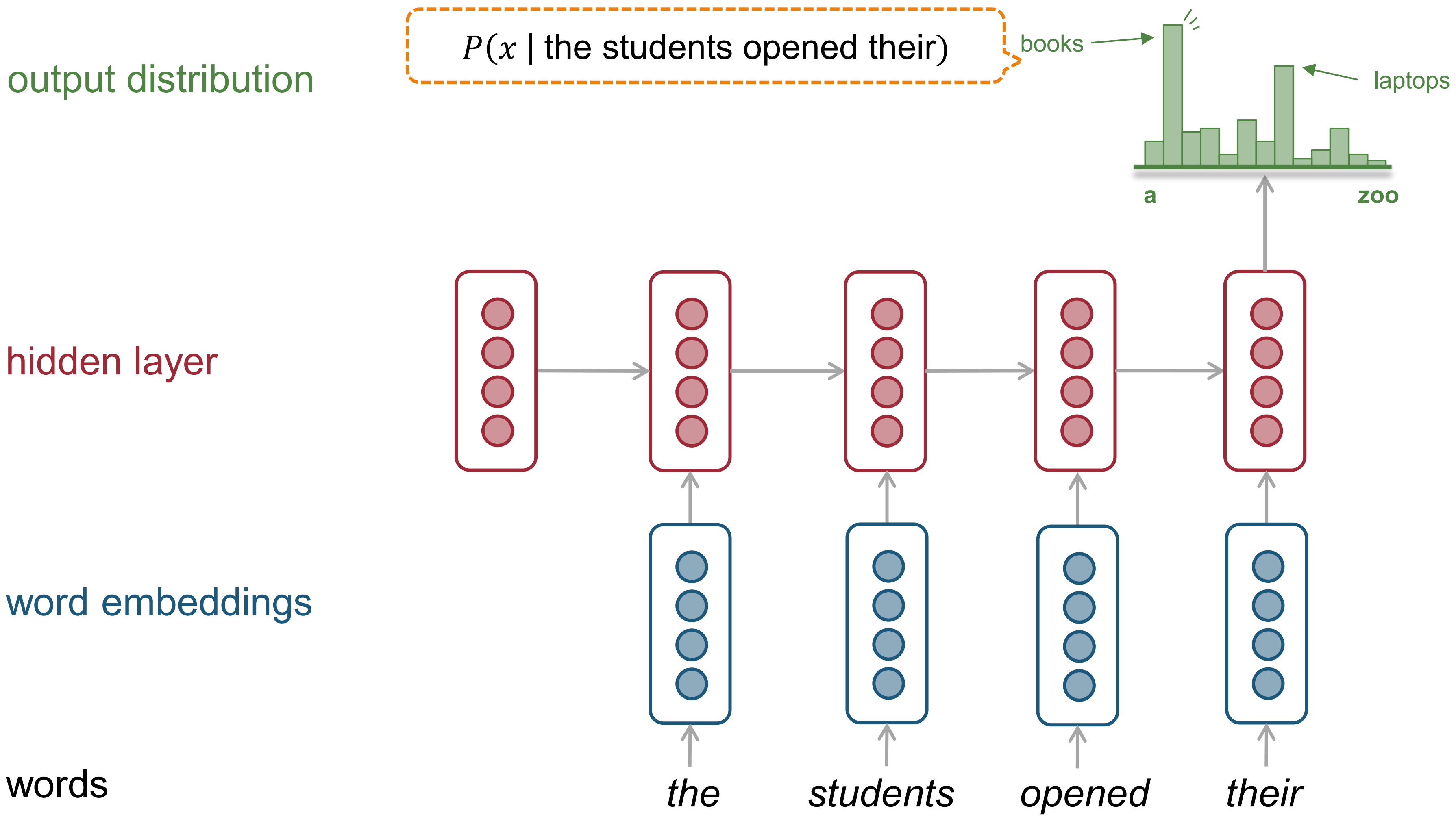
Hidden Markov Models (1960s)

- Each word has a hidden state (features)
  - An abstraction from words
  - Benefits: polysemy, synonym, …
- Markov model over hidden states



- Inference by variable elimination (forward algorithm)

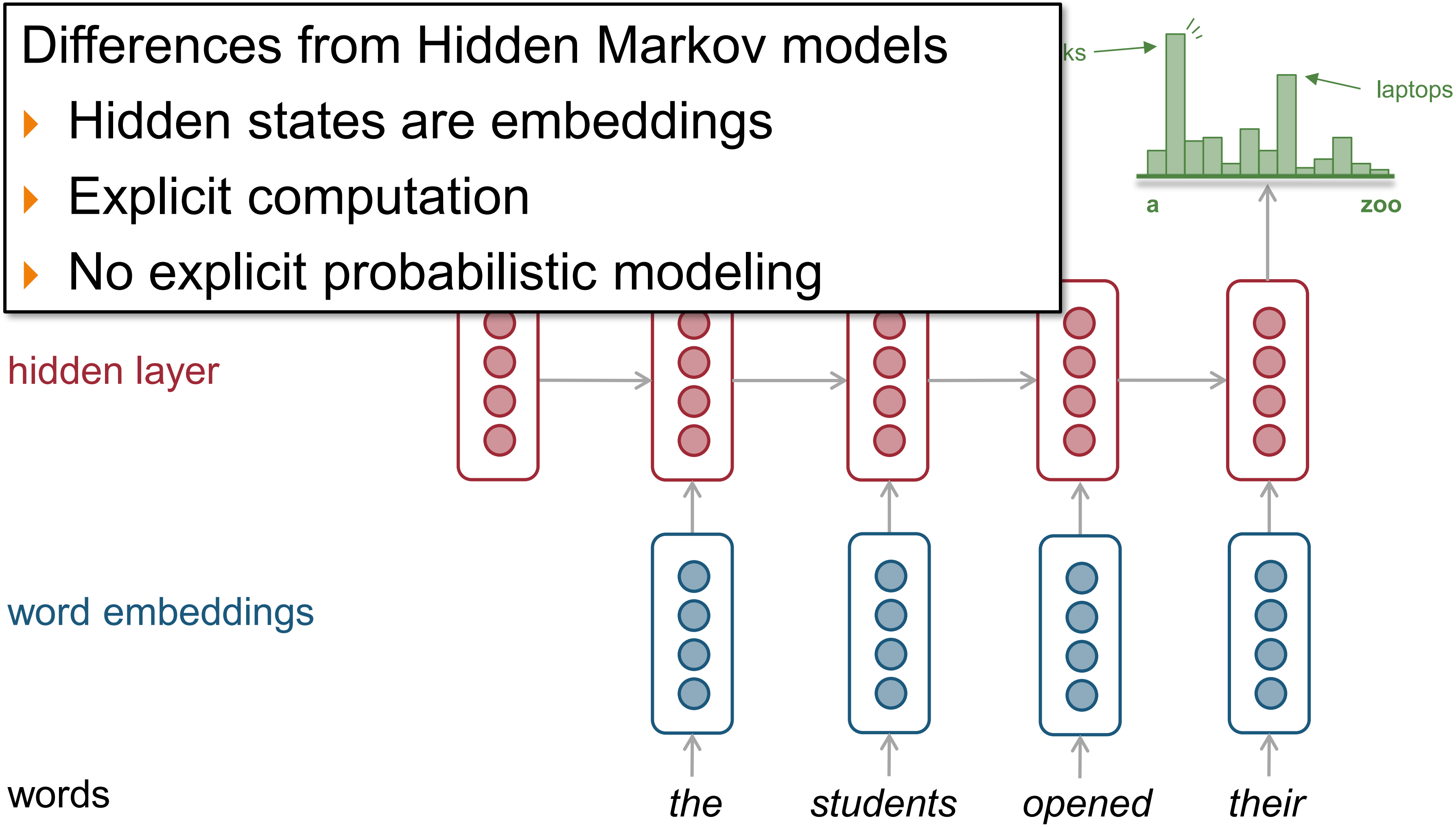
Recurrent neural network LM (1980s, 2010s)



Recurrent neural network LM (1980s, 2010s)

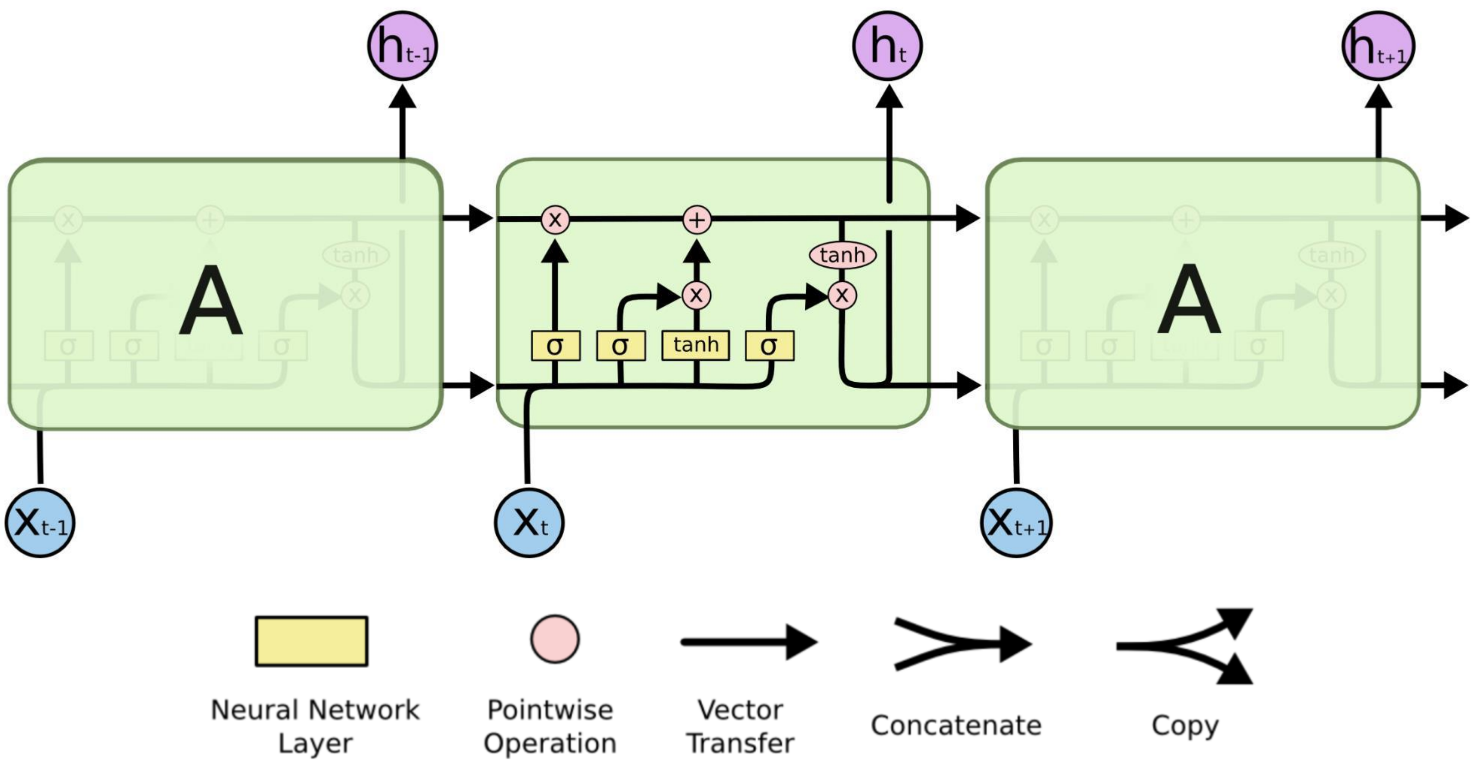
Differences from Hidden Markov models

- Hidden states are embeddings
- Explicit computation
- No explicit probabilistic modeling



Recurrent neural network LM (1980s, 2010s)

- More advanced variant: Long Short Term Memory (LSTM)
  - Three corresponding gates control the selection of which information is erased/written/read



Source: http://colah.github.io/posts/2015-08-Understanding-LSTMs/

Generating with an RNN LM

- RNN LM trained on Obama speeches:

The United States will step up to the cost of a new challenges of the American people that will share the fact that we created the problem. They were attacked and so that they have to say that all the task of the final days of war that I will not be able to get this done.

Source: https://medium.com/@samim/obama-rnn-machine-generated-political-speeches-c8abd18a2ea0

Problems with n-gram & RNN

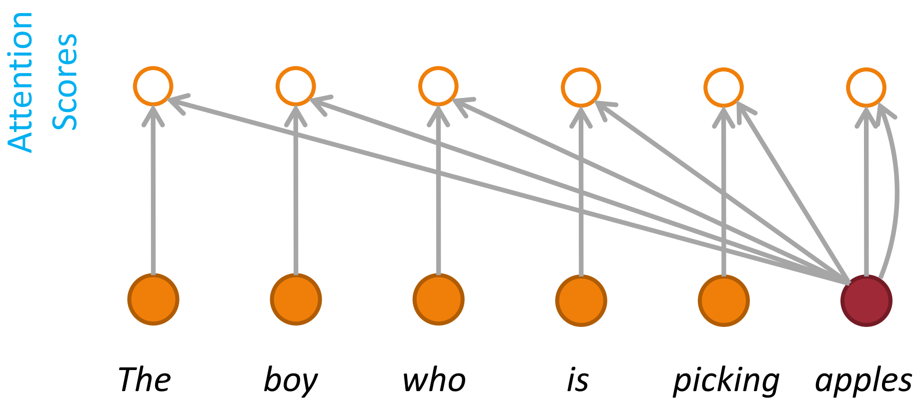
- *The boy who is picking apples* ____ is? are?

- N-gram LM & RNN not good at modeling long-range dependencies in natural languages.

Attention (2014)

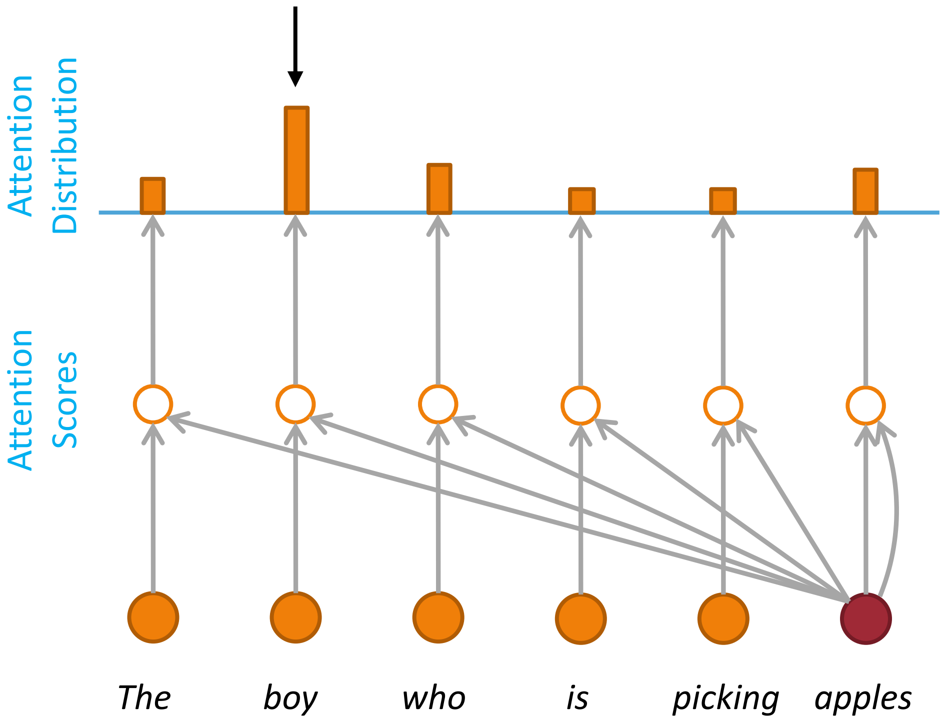
- At each step, add direct connections to a particular part of the history
  - Ex. attending “boy” when generating “is”
- But how do we know which part to attend to?
  - Attend to everything in the history
  - But learn to predict a different attention score for each word

Attention



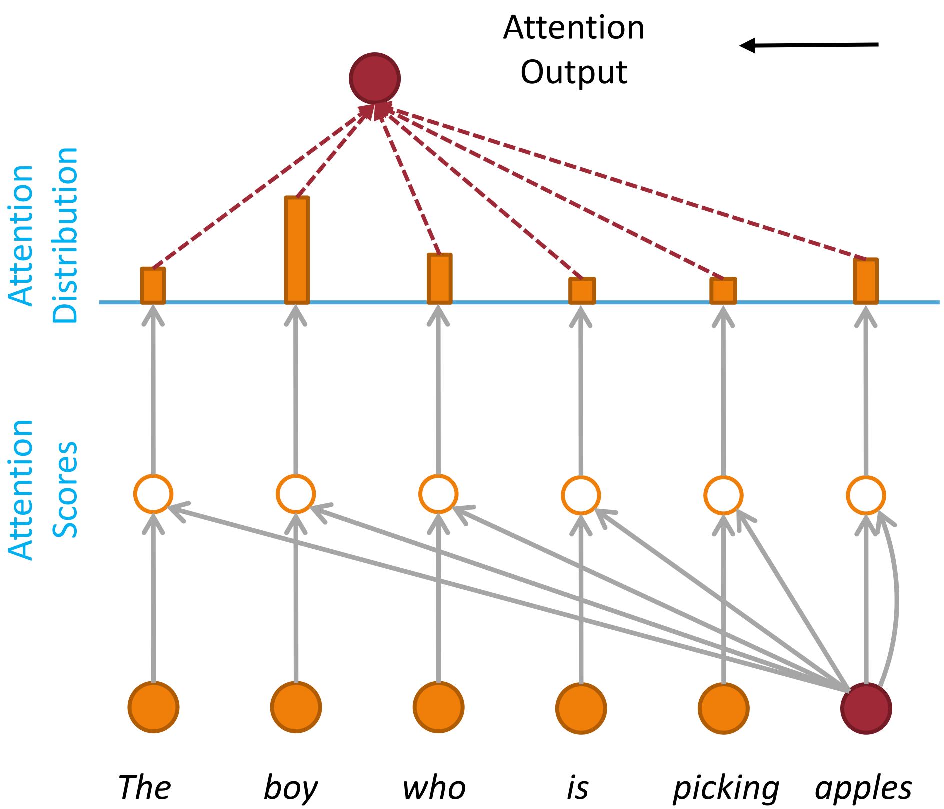
Attention

When generating “**is**”, we’re mostly focusing on the hidden state of “**boy**”



Apply **softmax** to turn the scores into a **probability** distribution

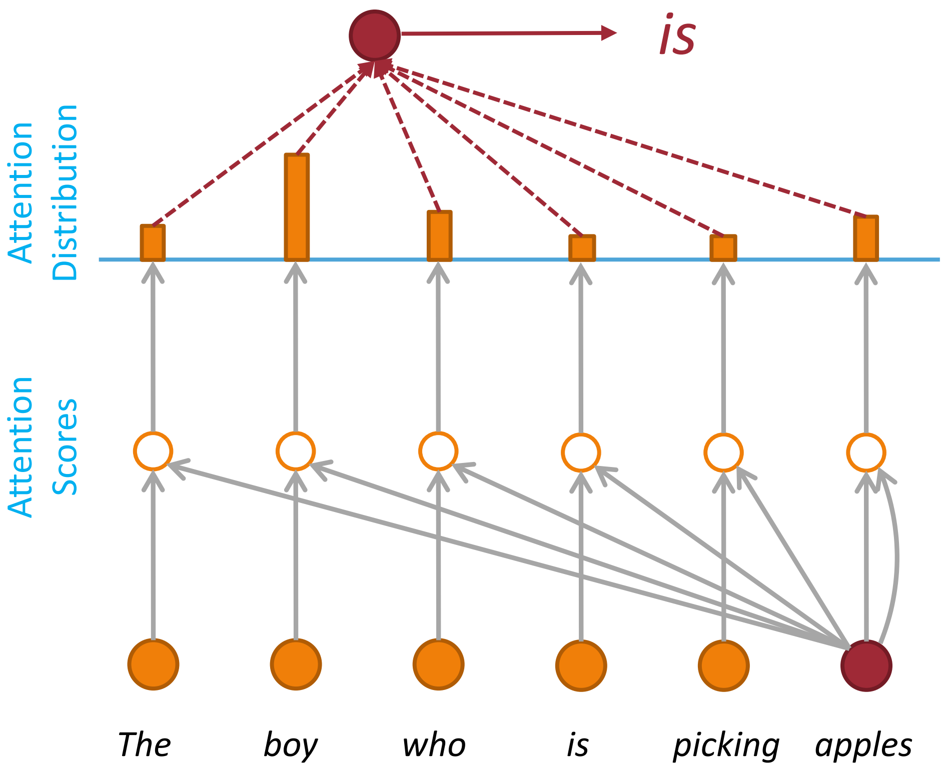
Attention



Use the attention distribution to take a weighted sum of all seen hidden states.

The attention output mostly contains information from the hidden states that received high attention.

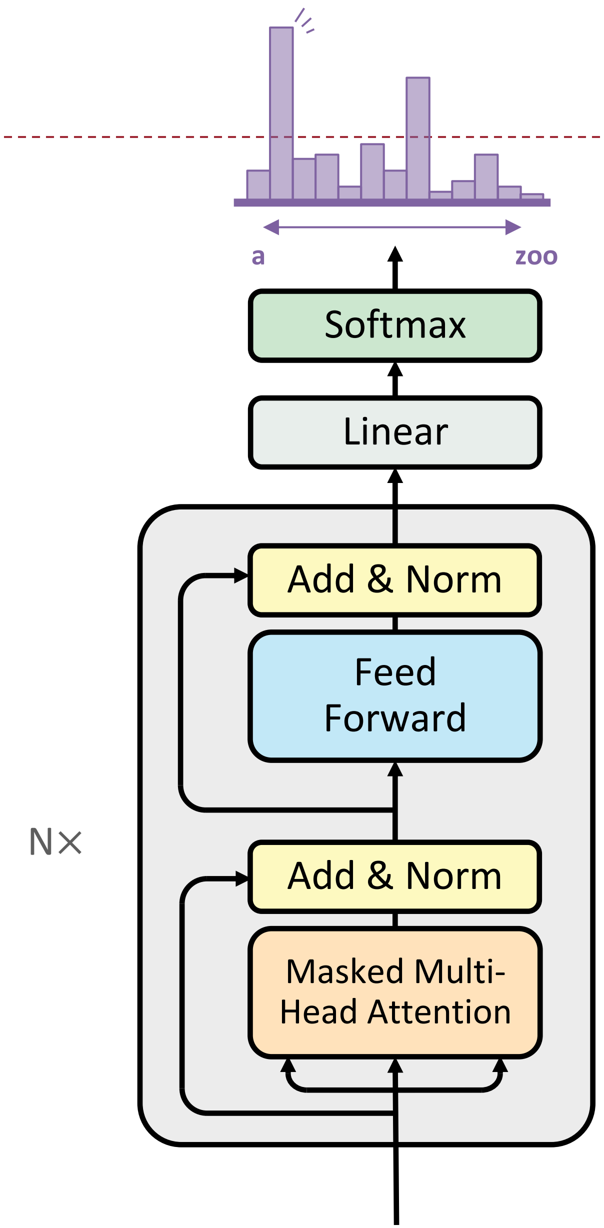
Attention



Use the attention output to predict the next word

Transformer (2017)

- Attention is all you need!
  - Not really…
- Additional components
  - Position embedding
  - Feedforward layer
  - Residual connection
  - Layer normalization
  - Stack multiple layers
- This is the basis of GPT & LLM



Generating with GPT-2

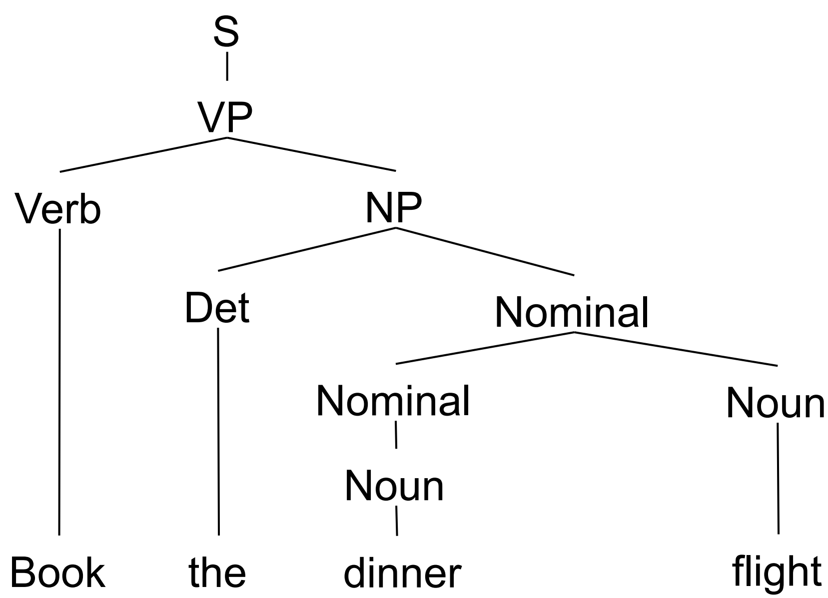
- Human prompt
  - In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.
- Model completion
  - The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

    Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.

    Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow.

Symbolism?

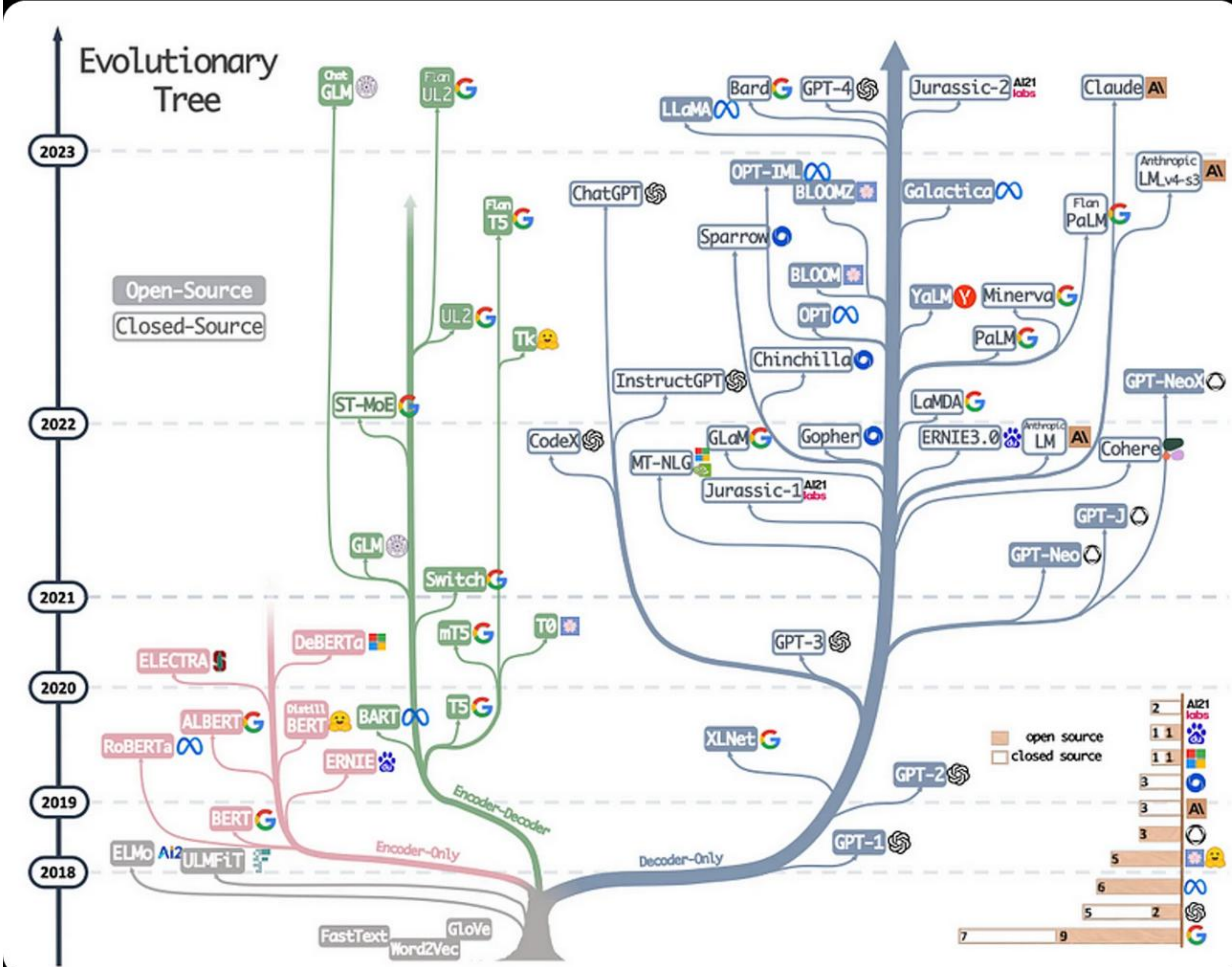
- Symbolism doesn't model probabilities.
  - A sentence is either correct or incorrect.
- Grammar: rule-based modeling of text
- We may add probabilities for language modeling



$S \rightarrow NP\ VP$
$S \rightarrow Aux\ NP\ VP$
$S \rightarrow VP$
$NP \rightarrow Pronoun$
$NP \rightarrow Proper\text{-}Noun$
$NP \rightarrow Det\ Nominal$
$NP \rightarrow Nominal$
$Nominal \rightarrow Noun$
$Nominal \rightarrow Nominal\ Noun$
$Nominal \rightarrow Nominal\ PP$
$VP \rightarrow Verb$
$VP \rightarrow Verb\ NP$
$VP \rightarrow Verb\ NP\ PP$
$VP \rightarrow Verb\ PP$
$VP \rightarrow Verb\ NP\ NP$
$VP \rightarrow VP\ PP$
$PP \rightarrow Preposition\ NP$

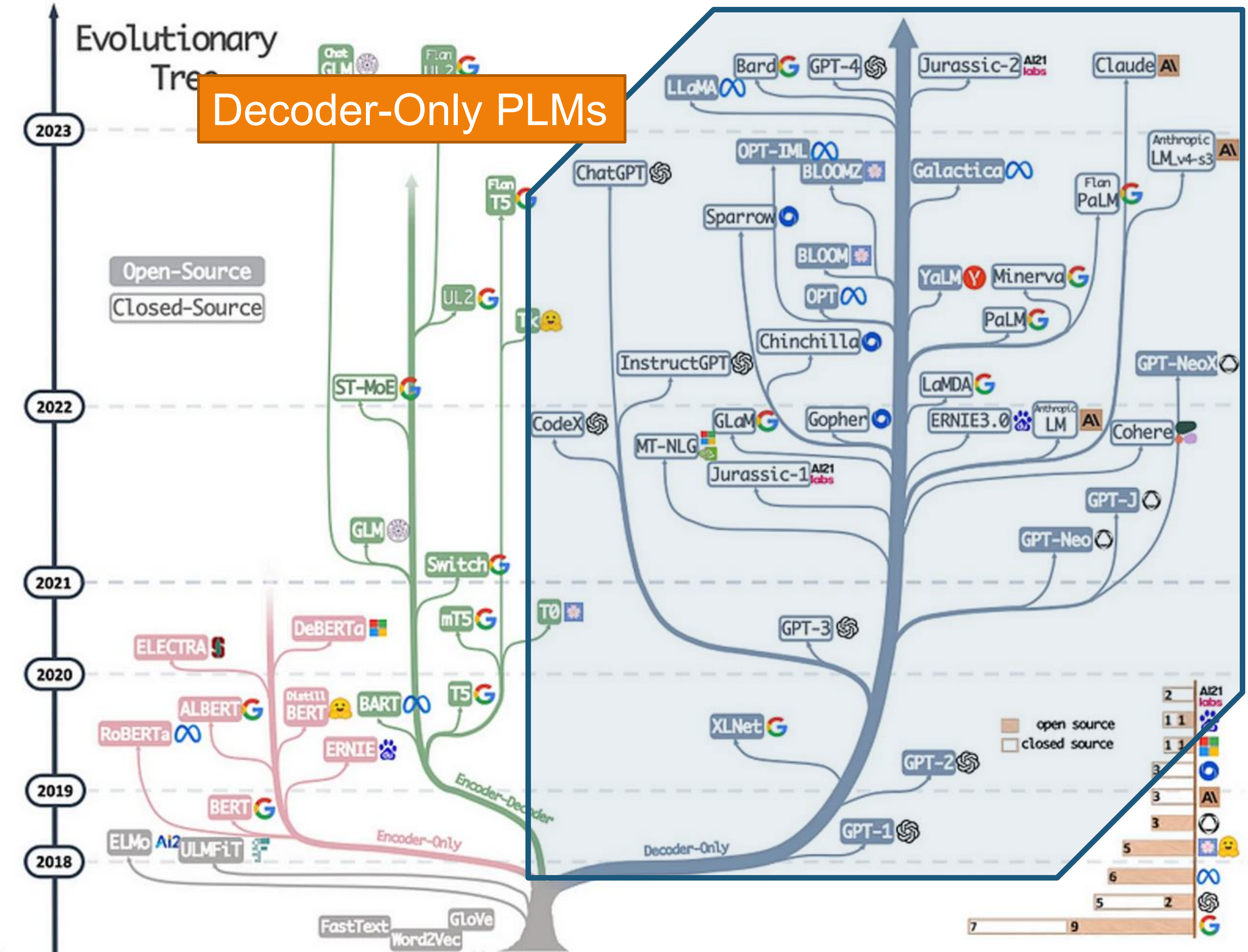
PLM & LLM

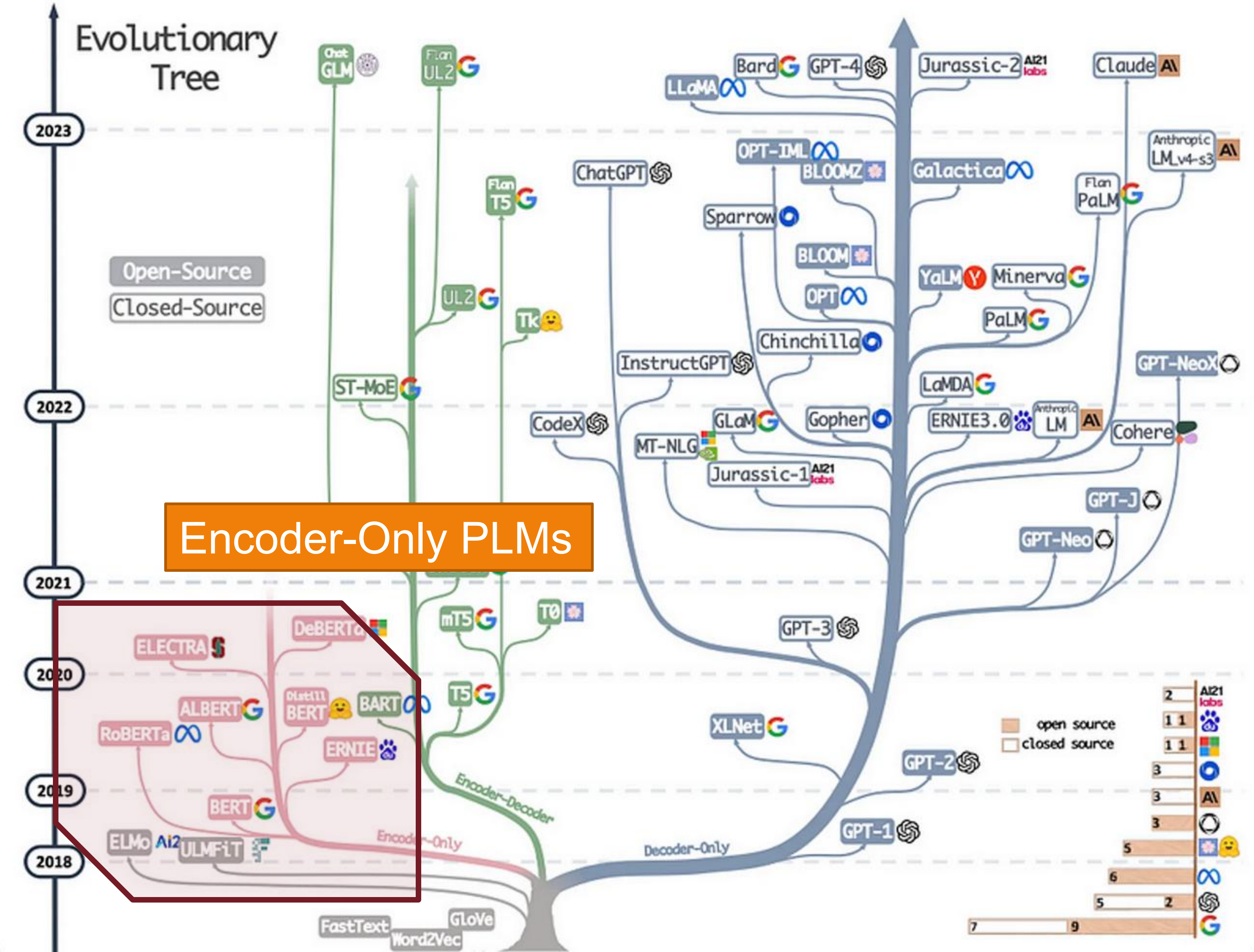
- Language model
  - $P(w_1, w_2, w_3, \cdots, w_n)$
- Pretrained language model
  - Train a general-purpose language model on a large corpus
  - Apply it to a task by finetuning it on task-specific data
- Large language model
  - Default definition: a language model (or variants) that is large (w.r.t. model size and training data size).
  - Current perception: a language model that does tasks by talking to it.



Evolutionary Tree
2023
2022
2021
2020
2019
2018
Open-Source
Closed-Source
Encoder-Only
Encoder-Decoder
Decoder-Only
FastText
Word2Vec
GloVe
ELMo
ULMFiT
BERT
RoBERTa
ALBERT
ELECTRA
Distill BERT
ERNIE
DeBERTa
BART
T5
mT5
T0
Switch
GLM
ST-MoE
UL2
Tk
Flan T5
Flan UL2
Chat GLM
GPT-1
GPT-2
XLNet
GPT-3
GPT-Neo
GPT-J
GPT-NeoX
CodeX
MT-NLG
Jurassic-1
GLaM
Gopher
ERNIE3.0
Anthropic LM
Cohere
LaMDA
InstructGPT
Chinchilla
OPT
BLOOM
Sparrow
PaLM
YaLM
Minerva
Flan PaLM
ChatGPT
OPT-IML
BLOOMZ
Galactica
Anthropic LM v4-s3
LLaMA
Bard
GPT-4
Jurassic-2
Claude
open source
closed source

Decoder-Only PLMs




Evolutionary Tree
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ChatGLM
Flan UL2
Flan T5
UL2
Tk
ST-MoE
mT5
T0
T5
BART
DeBERTa
ELECTRA
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DistillBERT
ERNIE
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Flan PaLM
Sparrow
BLOOM
YaLM
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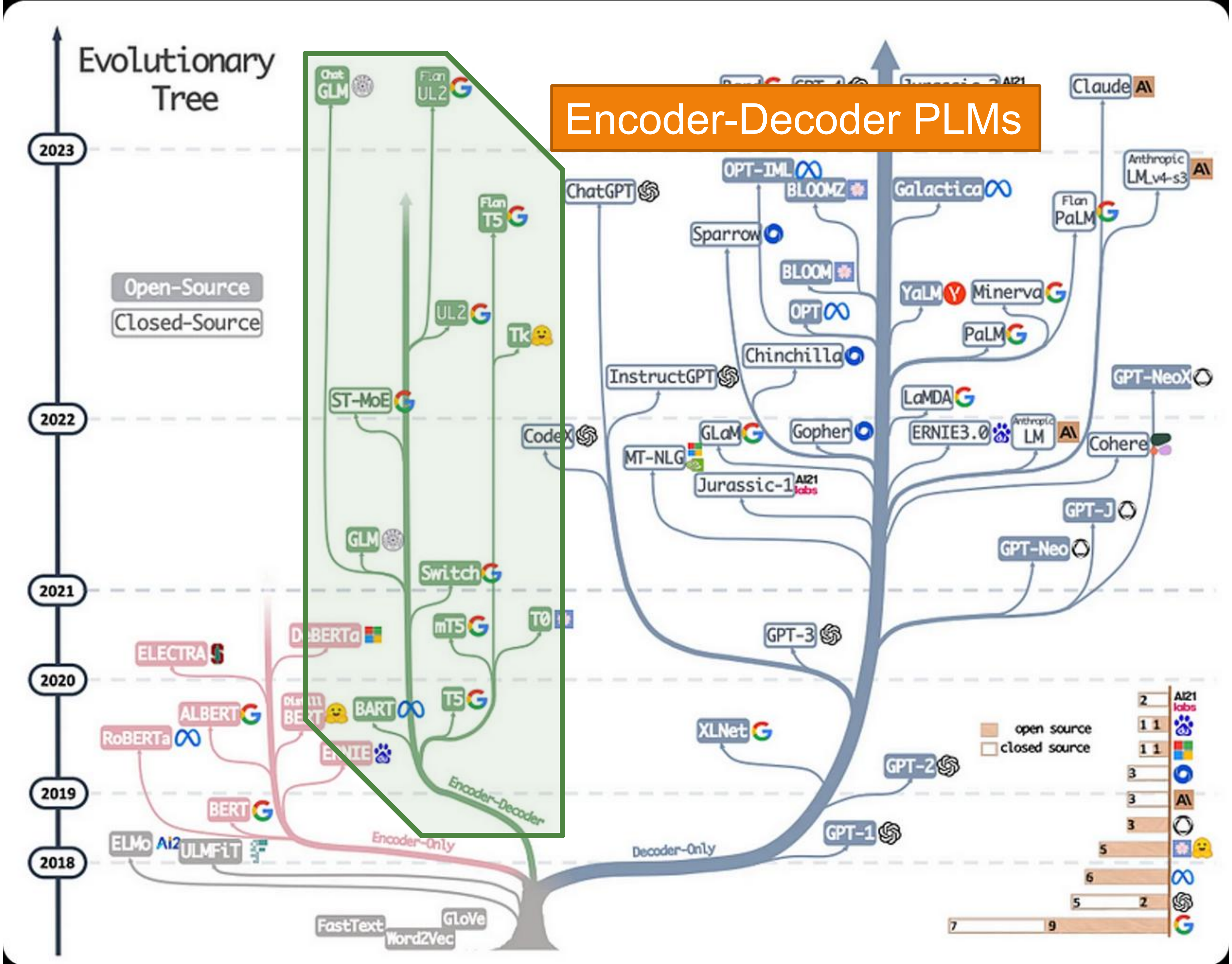
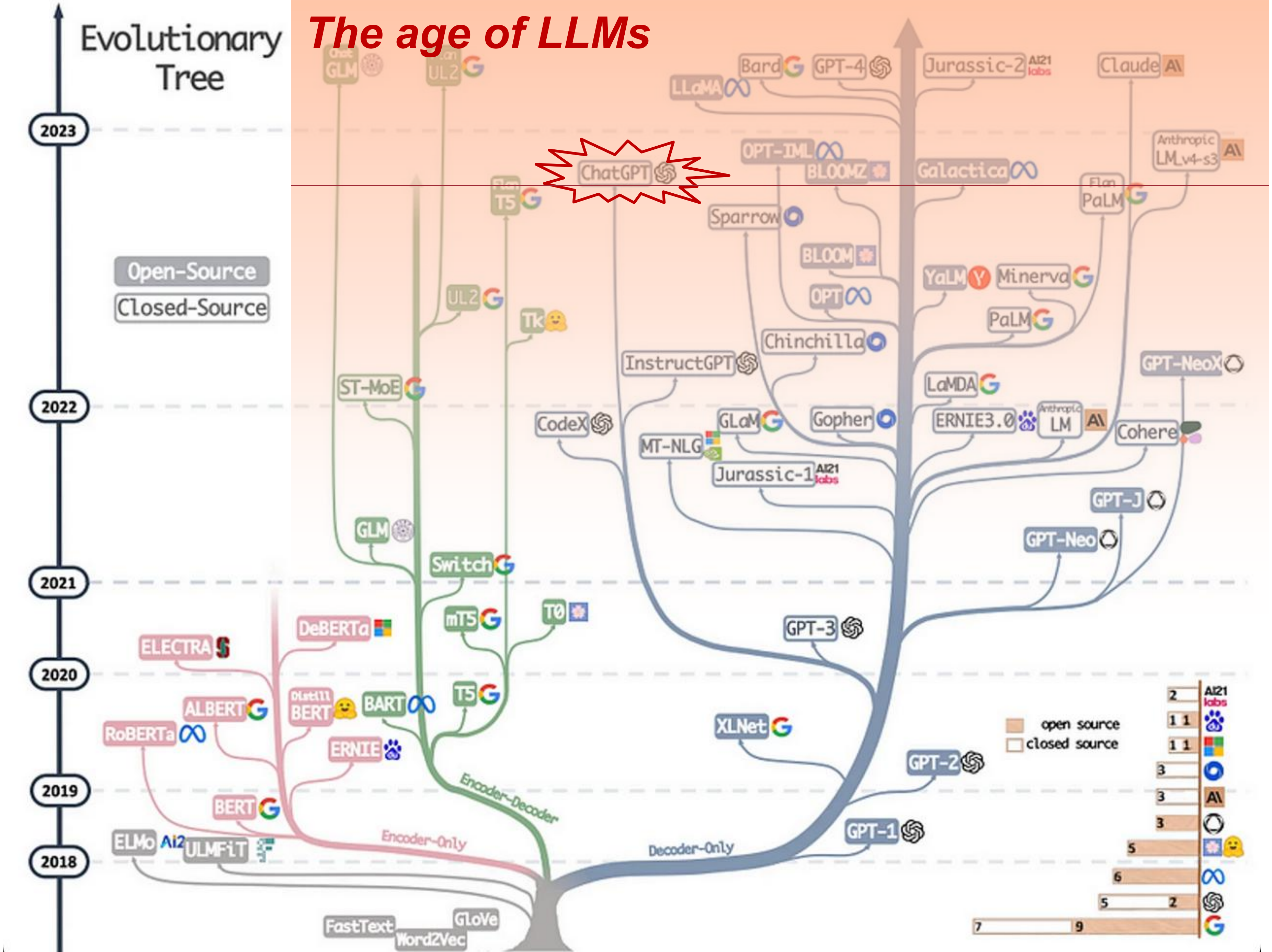
MT-NLG
Jurassic-1
GPT-J
GPT-Neo
GPT-3
XLNet
GPT-2
GPT-1
open source
closed source

Encoder-Decoder PLMs

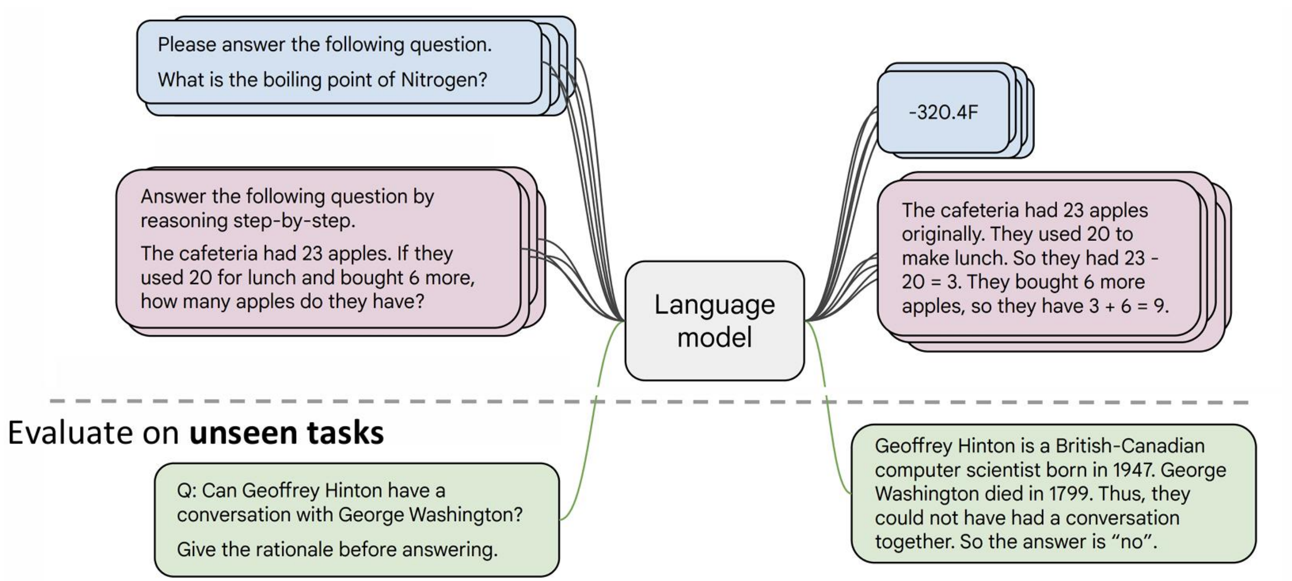


The age of LLMs



Training of an LLM

- Start from a pretrained decoder-only LM
  - Ex: GPT
- Instruction finetuning
  - We want the model to follow human instructions
    - Learning from QA pairs of tens of thousands of tasks



Training of an LLM

- Start from a pretrained decoder-only LM
  - Ex: GPT
- Instruction finetuning
  - We want the model to follow human instructions
    - Learning from QA pairs of tens of thousands of tasks
- Reinforcement learning with human feedback
  - We want the model to align with human preference
    - Learning from human ratings of model-generated responses

Large Reasoning Models

- LLM with reasoning ability
  - Ex: DeepSeek R1
- Reinforcement learning with outcome reward
  - Math: matching answer or not
  - Code: passing test cases
- Reinforcement learning with process reward
  - Ongoing research …

LLM – the good & the bad

- Language generation
- Language understanding
  - Instruction following
- World knowledge
- In-context learning
- Reasoning
- …

- abcde -> edcba
  acde -> edca
  deacd -> dcaed
  bdac ->
- cadb

- The bad?

ChatGPT @2023.3

- Factual errors (hallucination)
  - 请介绍一下加州大学的哈利波特
  - 加州大学的“哈利波特”是指该校的伯克利分校（University of California, Berkeley）图书馆里收藏的一本古老的魔法书，正式名称为《The Book of Magic》。这本书据称是15世纪意大利一位神秘的作者写成的，包含了各种魔法、咒语、药水等方面的内容。

    这本书在伯克利分校的罕见书籍收藏中极为珍贵，因为它是唯一一本完整的手抄本，且保存完好。该书通常存放在伯克利分校的梅恩图书馆（The Bancroft Library）中，由于价值非凡，只有特殊的许可证才能查看。

    该书之所以被称为“哈利波特”，是因为它在2000年左右被发现的时候，正值J.K.罗琳的“哈利波特”系列小说风靡全球的时期，而这本古老的魔法书被认为是“真正的哈利波特”。因此，这个昵称就一直沿用至今。

ChatGPT @2023.3

- Factual errors (hallucination)
  - 请介绍一下上海科技大学的哈利波特
  - 上海科技大学的“哈利波特”是指该校的一位学生，名叫李[redacted]，他在2019年发表的一篇论文被誉为“计算机视觉领域的哈利波特”。

    这篇论文题为《MUNIT: Multimodal Unsupervised Image-to-Image Translation》，是在他就读于上海科技大学时完成的。这篇论文提出了一种新的图像转换方法，能
    风格，例如将夏天的风景转换为
    马的图像等等。

    这种方法被称为多模态无监督图
    Unsupervised Image-to-Image
    点是不需要人工标注的训练数据，
    系，并生成高质量的转换结果。

    这篇论文发表后引起了广泛关注，被认为是计算机视觉领域的一项重大突破，因此李[redacted]也被誉为“计算机视觉领域的哈利波特”。

Multimodal Unsupervised Image-to-Image Translation

Xun Huang[1], Ming-Yu Liu[2], Serge Belongie[1], Jan Kautz[2]

Cornell University[1] NVIDIA[2]

ChatGPT @2024.5

- 请介绍一下上海科技大学的哈利波特
- 上海科技大学（ShanghaiTech University）也没有与哈利波特相关的特定版本或项目。哈利波特是由英国作家J.K. 罗琳（J.K. Rowling）创作的一系列奇幻小说，描述了年轻巫师哈利·波特在霍格沃茨魔法学校的冒险故事。这些小说已被翻译成多种语言，并且在全球范围内享有极高的知名度和广泛的读者群。

  然而，上海科技大学的学生和教职员工可能会在校园内举办各种与哈利波特相关的活动，比如阅读俱乐部、电影放映、主题派对等，以庆祝这个受欢迎的文化现象。这些活动通常是由学生社团或校园组织自发组织的，而非学校官方举办的。

ChatGPT @2024.5　　Let’s try again…and again…

- 请介绍一下上海科技大学的哈利波特
- 上海科技大学的哈利波特活动是该校的一项传统活动，灵感来源于J.K.罗琳的哈利波特系列。在这个活动中，学生们会模仿哈利波特系列中的情节和角色，穿着类似哈利波特系列中角色的服装，例如学院制服、巫师帽等。他们会在校园内举行各种有趣的活动，例如模拟魔法课、魔药课和飞行课等。此外，还可能包括表演、竞赛和手工艺活动等。这个活动旨在为学生提供一个快乐的、富有创意的社交活动，同时也是展示他们想象力和创造力的机会。通过参与这样的活动，学生们可以加强彼此之间的联系，共同享受哈利波特系列带来的乐趣。

ChatGPT @2024.5

- 请介绍一下上海科技大学医学院的李时珍老师
- 上海科技大学医学院的李时珍老师是一位备受尊敬的医学教育者和学者。李时珍老师在医学领域有着丰富的教学和研究经验，尤其在中药学方面有着深厚的造诣。

  李时珍老师可能是以其在中医药学方面的专业知识和贡献而闻名。作为一名教育者，他可能致力于教授学生中药学的理论知识、临床应用以及药材鉴定等方面的内容。同时，作为一位研究者，他可能在中药药理、药效学、中药质量控制等领域进行了深入的研究工作，为中医药的发展和传承做出了重要贡献。

  李时珍老师可能以其对中医药的热爱和执着，以及对学生的悉心教导而备受学生和同行的尊敬和爱戴。他可能在教学和科研工作中注重理论与实践相结合，致力于培养具备扎实专业知识和临床技能的医学人才，为中医药事业的发展和传承作出了重要贡献。

ChatGPT @2023.3

- Incapable of complex reasoning
  - 请证明根号3是无理数
  - (A correct and very detailed proof)
  - 同理，请证明根号4是无理数
  - 我们同样使用反证法来证明根号4是无理数

    …

    假设根号4是有理数，则存在两个整数p和q（q ≠ 0，且p、q互质），使得：根号4 = p/q

    …

    (2k)^2 = 4q^2

    4k^2 = q^2

    …

    因此，p和q有一个共同的因子2，与我们最初的假设矛盾，即p和q互质。…

ChatGPT @2024.5

- 请证明根号3是无理数
- (A correct and very detailed proof)
- 同理，请证明根号4是无理数
- 证明根号4是无理数其实是不必要的，因为根号4等于2，而2是一个有理数。有理数是可以表示为两个整数的比值，而2可以表示为2/1，因此2是一个有理数。因此，根号4是有理数，而不是无理数。

  如果你想了解无理数的例子，可以考虑根号2。根号2无法表示为两个整数的比值，因此是一个无理数。

ChatGPT @2024.5

- 同理，请证明根号289是无理数
- 我们可以进行同样的间接证明法来证明$\sqrt{289}$是无理数。

  假设$\sqrt{289}$是有理数，可以表示为一个不可约分数$\frac{a}{b}$，其中a和b是整数，并且互质。
  ...
  整理得到：289*k^2 = b^2
  这表明b^2也是289的倍数，因此b也是289的倍数。

  然而，这与假设中a和b互质的前提相矛盾。因此，假设$\sqrt{289}$是有理数导致了矛盾，我们得出结论：$\sqrt{289}$是无理数。

ChatGPT @2023.3

- Arithmetic
  - 10242048-256512=?
  - 10242048-256512=9985504

  - 请生成一个包含十个汉字的句子
  - 秋天的落叶缓缓飘落在湖面上，像是一幅精美的水墨画。
  - 这句话不止十个汉字啊
  - 非常抱歉，我的计数出现了错误。以下是一个包含十个汉字的句子：
    花开花落，岁月流转，世事无常。

ChatGPT @2024.5

- Arithmetic
  - 10242048-256512=?
  - The result of subtracting 256512 from 10242048 is 9985504.

  - 请生成一个包含十个汉字的句子
  - 当夜幕降临，星光点亮了天空，大地安静无声。
  - 这句话不止十个汉字啊
  - 抱歉，我理解错了。这是一个包含十个汉字的句子：夜幕降临，星光点亮了天空。

GPT-4 @2024.5

- 10242048-256512=?
- 10242048 - 256512 = 9979536

- 请生成一个包含十个汉字的句子
- 好的，以下是一个包含十个汉字的句子：“春暖花开，百花齐放。”
- 标点也算汉字吗？
- 不包括标点符号在内，一个汉字通常由一个或多个汉字字符组成。

ChatGPT @2024.5

- Lack common sense
  - 现在有一个袋子，如果我们先将一个苹果放入的这个袋子中，然后又从这个袋子中取出了一个橘子，那么在苹果放到这个袋子之前，袋子中有什么？
  - 在苹果放入袋子之前，袋子内是空的，里面什么都没有。

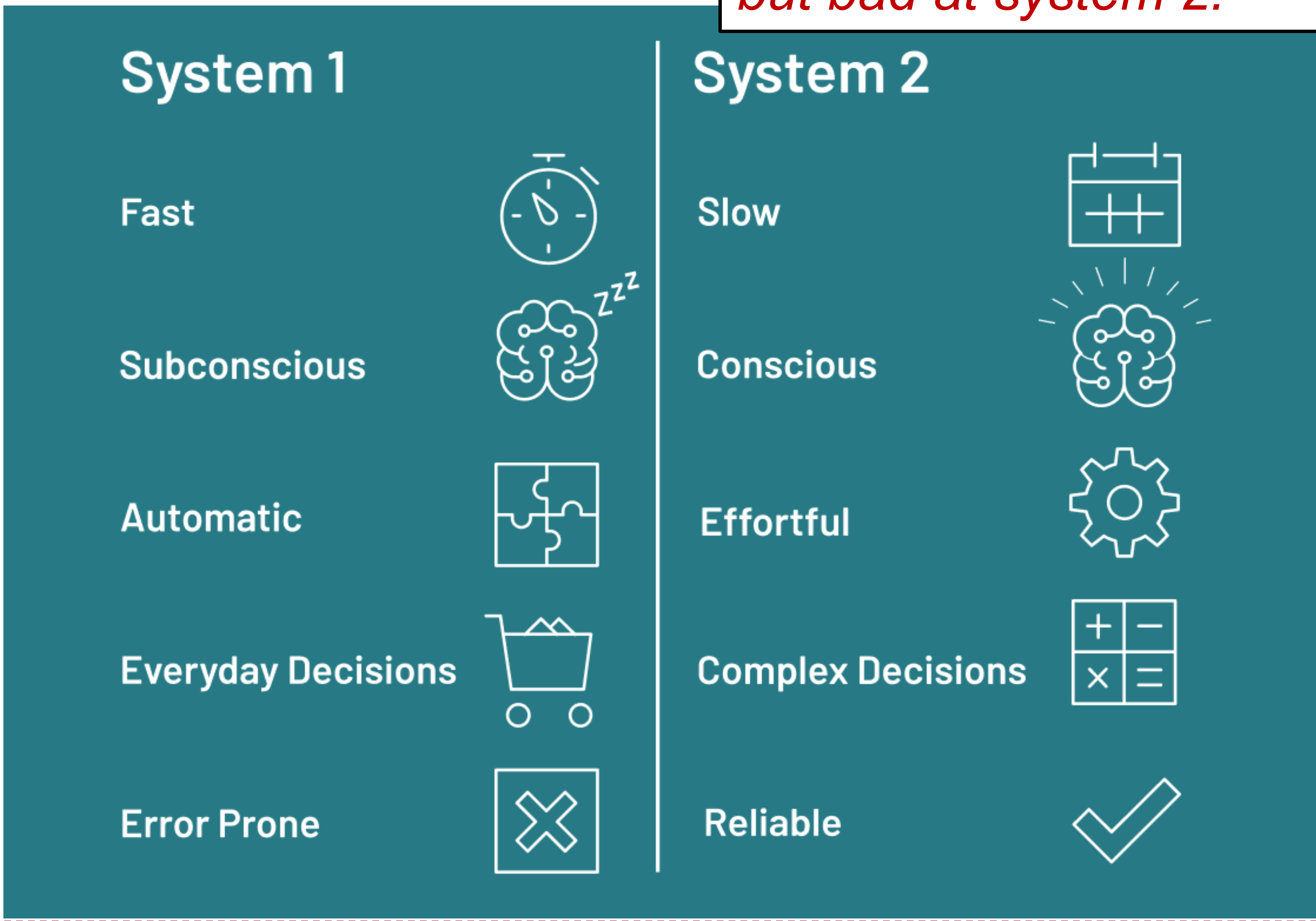
What LLMs are not good at

- Factual errors (hallucination)
  - Esp. corner cases
- Incapable of complex formal reasoning
- Bad at arithmetic
- Lack common sense

- What do they have in common?

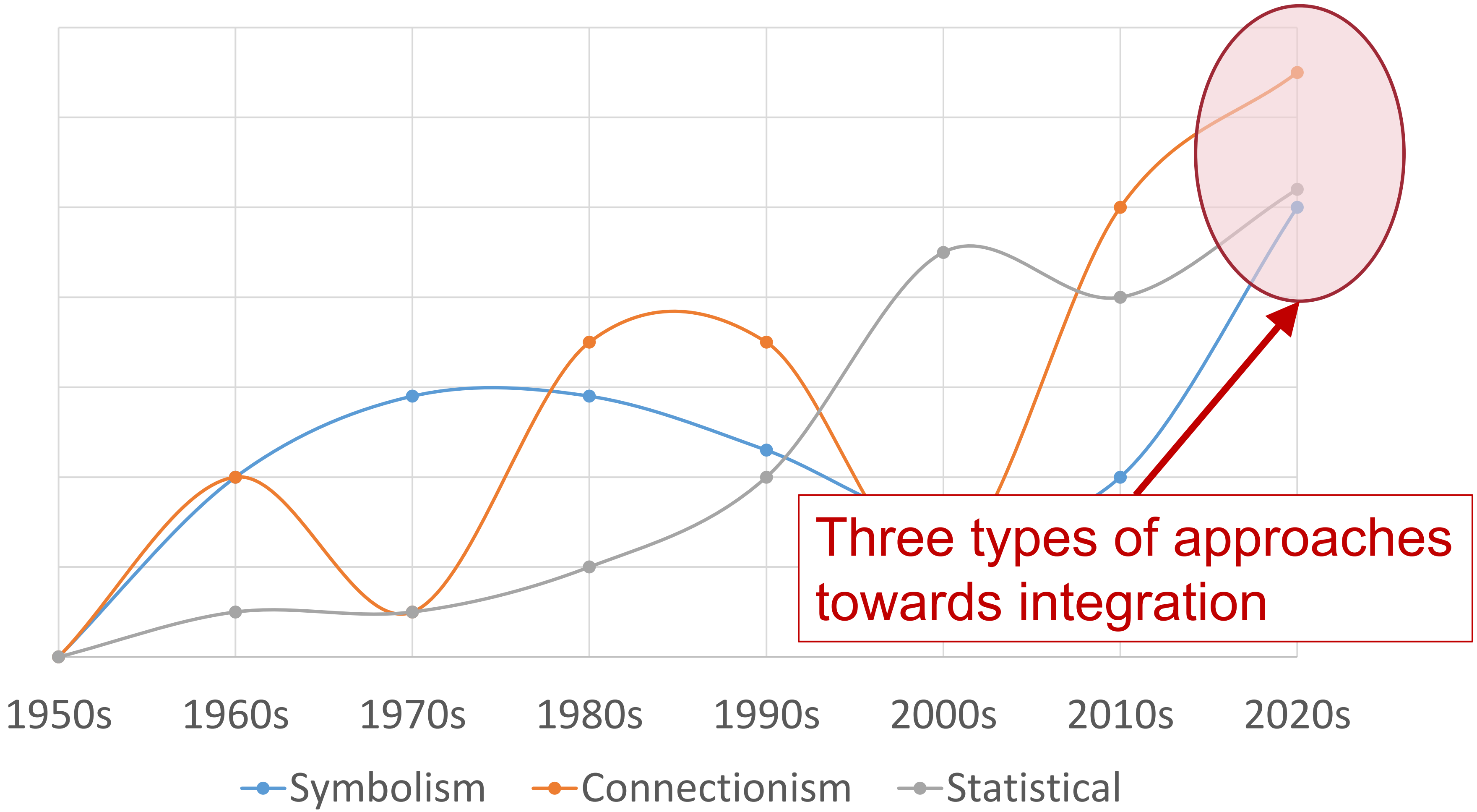
System 1 vs. System 2

LLMs are good at system 1, but bad at system 2!



https://neurofied.com/thinking-fast-slow-down-system-1-and-2/

Trends


Three types of approaches towards integration
1950s
1960s
1970s
1980s
1990s
2000s
2010s
2020s
Symbolism
Connectionism
Statistical